

The Limits of Falsifiability: Dimensionality, Measurement Thresholds, and the Sub-Landauer Domain in Biological Systems

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Abstract

Scientific inquiry repeatedly compresses rich observations into decision boundaries (accept/reject, significant/non-significant, model A/model B). We argue that this compression is *scale-dependent* and can *saturate* in high-dimensional biological systems operating near fundamental measurement limits. But the limitation runs deeper than measurement: *falsifiability itself depends on axiomatic choices about how questions are structured*, and these framework choices are themselves projections that cannot be tested from within the framework. In neural networks, much relevant information exists as patterns below reliable single-event readout energy (with $k_B T \ln 2$ as reference scale)—signals that no single detector can reliably read, yet detectable when pooled

across populations. We do *not* claim such phenomena are unmeasurable in aggregate; we claim *single-event bit-level readout* is unreliable, so only ensemble/statistical observables are accessible. These sub-threshold patterns cannot be projected into binary outcomes without potentially destroying their functionally relevant causal structure. We develop a framework connecting dimensionality, thermodynamic constraints on reliable readout, framework dependence, and biological epistemology, showing that Popperian-style demarcation is most effective in low-dimensional systems with strong signals *and* shared axiomatic structure. The “unreasonable effectiveness of mathematics” in physics reflects selection bias toward domains where projection loss is small; biology is where this selection breaks down. We propose a scale-dependent falsifiability criterion (Section 9) and motivate a shift from single-case hypothesis tests to multi-scale, ensemble-based inference that makes framework assumptions explicit.

Keywords: falsifiability, dimensionality, Landauer limit, framework dependence, Duhem-Quine thesis, stochastic resonance, consciousness, biological epistemology

1 Introduction

Karl Popper’s falsifiability criterion has served as a cornerstone of scientific epistemology since its articulation in *The Logic of Scientific Discovery* [1]. The framework’s elegance lies in its binary clarity: a hypothesis is scientific if and only if it can, in principle, be shown false through empirical observation. This criterion has proven remarkably effective in distinguishing science from pseudoscience across biology, from molecular genetics to ecosystem ecology.

However, recent advances in our understanding of biological complexity and physical measurement limits expose fundamental boundaries to Popper’s framework. Complex biological systems, ranging from neural networks to protein folding landscapes, operate in high-dimensional phase spaces where causal relationships emerge from the collective behavior of many interacting components. We argue that falsifiability, rather than being a uni-

versal criterion for biological knowledge, applies only to systems meeting specific physical, mathematical, *and axiomatic* constraints.

1.1 Three Levels of Limitation

This paper identifies three distinct levels at which falsifiability breaks down:

1. **Physical measurement limits.** The Landauer principle sets a lower bound on irreversible bit erasure ($\sim k_B T \ln 2$); in biological settings it serves as a reference scale for reliable single-event discrimination under thermal noise [2]. Many biological patterns exist below this threshold—causally potent yet not reliably readable as individual bits.
2. **Dimensional projection loss.** High-dimensional systems projected onto low-dimensional observations lose almost all information. A binary test on a 100-neuron circuit preserves less than 1% of the information content.
3. **Framework dependence.** Before any measurement occurs, the choice of what counts as a test, what counts as evidence, and how the question is structured already embeds unfalsifiable assumptions. The framework is itself a projection.

The third level is the deepest and most often overlooked. Even in domains where measurement is possible and dimensionality is manageable, falsifiability remains relative to a framework that cannot itself be falsified from within.

Notation and definitions. We use the following throughout:

- X : System state space (high-dimensional)
- Y : Observation space (raw measurements)
- Z : Representation space (processed statistics, model outputs)
- $\mathcal{F} = (M, R, D)$: A *framework*, where $M : X \rightarrow Y$ is measurement, $R : Y \rightarrow Z$ is representation, and $D : Z \rightarrow \{0, 1\}$ is demarcation

- *Projection loss*: $\mathcal{L}(\mathcal{F}) := H(X) - I(X; Z)$, the information about X that is unrecoverable after passing through the pipeline $X \rightarrow Y \rightarrow Z$. By the data-processing inequality, $I(X; Z) \leq I(X; Y) \leq H(X)$; each stage cannot increase recoverable information about X . We use this quantity qualitatively; no rigorous cross-domain quantification exists.
- *Binary epistemology*: Any methodology that forces partitions on X (accept/reject, significant/non-significant)

1.2 The “Unreasonable Effectiveness” as Selection Bias

Eugene Wigner famously noted the “unreasonable effectiveness of mathematics in the natural sciences” [3]—the surprising correspondence between abstract mathematical structures and physical reality. We propose a partial explanation: physics has historically selected for systems where mathematical description works well.

Any *implemented* model has finite description length—finite parameters, finite precision, finite compute—and hence defines a finite-information projection of the system. Real systems, especially biological ones, have state spaces of enormous or effectively infinite dimension. Every mathematical model is already a projection, a shadow of the actual dynamics.

Physics appears successful partly through selection bias: we call “physics” the domains where projection loss is small enough for precise prediction—isolated systems, controlled conditions, symmetric situations where relevant degrees of freedom are few. The success of mathematics in physics does not demonstrate that mathematical description is universally adequate; it demonstrates that we have been studying the domains where it is adequate.

Biology is where projection loss becomes undeniable. Living systems maintain high-dimensional internal states that exceed observational access—not as a limitation of current technology, but as part of what makes them alive. The persistent difficulty of reducing biology to physics, the tendency of organisms to be “more than the sum of their parts,” may reflect a genuine feature of the subject matter rather than temporary limitation.

These considerations motivate a key result developed in Section 9: *scale-dependent falsifiability*. A hypothesis is falsifiable only when the energy scale of relevant patterns exceeds fundamental measurement thresholds, when the dimensionality is low enough for projection loss to be acceptable, *and* when framework assumptions are shared. Enzyme kinetics admits falsification; consciousness may not.

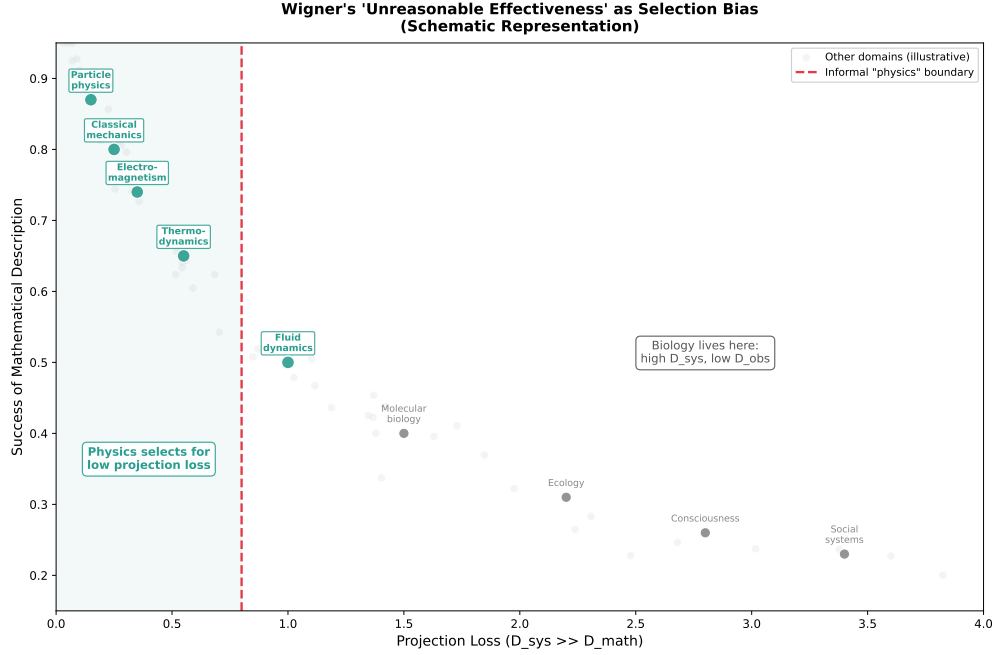


Figure 1: The “Unreasonable Effectiveness” as Selection Bias (Schematic Representation). Domains studied by physics (green) cluster where projection loss is small—where finite-dimensional mathematical descriptions capture most of the relevant dynamics. Biology, consciousness, and social systems occupy the high projection-loss regime where mathematical description fails. The apparent success of mathematics in physics reflects selection for tractable domains, not a deep truth about the mathematical nature of reality. *Note: Key domain positions are manually placed to illustrate the qualitative claim; no rigorous quantification of “projection loss” exists across these diverse fields. The figure visualizes the argument, not empirical measurements.*

2 Framework Dependence: The Deepest Limitation

2.1 The Duhem-Quine Insight Extended

Duhem [4] and Quine [5] established that no hypothesis is tested in isolation. Any empirical test involves auxiliary assumptions about measurement instruments, background conditions, and what counts as evidence. When a prediction fails, logic alone cannot determine whether the hypothesis or an auxiliary assumption is at fault.

We extend this insight: *the framework within which hypotheses are formulated is itself a projection*. Before you choose a hypothesis to test, you have already made choices about:

- What variables are relevant
- What counts as an observation
- What precision is sufficient
- How the question is structured
- What background knowledge is assumed

These choices constitute a dimensional reduction. The full space of possible framings is high-dimensional; any specific framing projects this into a particular low-dimensional subspace. Different researchers asking “different questions” are often occupying different projections of the same underlying reality.

2.2 Framework Choice as Dimensional Reduction

Consider an analogy. A three-dimensional object casts different shadows on different walls depending on the projection angle. Two observers seeing different shadows might disagree about the object’s shape—one sees a circle, another sees a rectangle—even though they are observing the same object.

Similarly, researchers in different disciplines, or even within the same discipline using different paradigms, are projecting high-dimensional reality onto different low-dimensional coordinate systems. Their disagreements may not be resolvable by evidence because they are not making claims in the same framework.

Definition (Framework). A *framework* $\mathcal{F}$ is a triple (M, R, D) mapping $X \rightarrow Y \rightarrow Z \rightarrow \{0, 1\}$, where $M : X \rightarrow Y$ is measurement, $R : Y \rightarrow Z$ is representation, and $D : Z \rightarrow \{0, 1\}$ is the demarcation judgment. Here X is the high-dimensional state space of the system, Y is the space of raw observations, Z is the space of processed representations (statistics, model outputs). Each map involves dimensional reduction: $\dim(Y) < \dim(X)$, $\dim(Z) < \dim(Y)$, and the final decision is one-dimensional. The total information loss compounds across stages. Framework dependence enters at every step: the choice of M (what to measure), the choice of R (how to represent), and the choice of D (where to draw the boundary).

Each choice of (M, R, D) induces a *partition* on X : states that map to the same final decision are equivalent under that framework. Two frameworks are *incompatible* when they induce different partitions, so “the same hypothesis” is not even well-defined across frameworks. Falsifiability is defined relative to $\mathcal{F}$, because a hypothesis is a statement about the induced partition.

This is not relativism. The underlying reality exists independently of the frameworks used to describe it. But *falsification is framework-relative*. A hypothesis can only be falsified with respect to a given set of background assumptions, and those assumptions cannot themselves be falsified within the framework that presupposes them.

2.3 Why Physics “Works”

Physics appears to escape this problem because its framework assumptions are unusually stable and widely shared. Physicists agree on what counts as a measurement, what the relevant variables are, and how precision is assessed. The axiomatic structure is settled.

But this stability is not solely a feature of nature; it also reflects disciplinary convergence and domain selection. Even within physics, the domains where mathematical description struggles—fully developed turbulence, strongly correlated many-body systems, climate dynamics—are precisely those with high effective dimensionality and contested framework choices. Physics has historically focused on domains where:

1. System dimensionality is low or effectively reducible
2. Measurement is clean and repeatable
3. Framework assumptions are uncontested
4. Projection loss is small

In domains where these conditions fail—consciousness, ecology, evolution, social systems—falsifiability becomes contested not because researchers are irrational but because they occupy genuinely different frameworks that project reality differently.

2.4 The Regress Problem

One might attempt to resolve framework disputes by stepping back to a meta-framework that adjudicates between frameworks. But this meta-framework is itself a projection, subject to the same limitations. The regress does not terminate.

This is not a counsel of despair. It is a recognition that empirical knowledge is always framework-relative, and that the framework itself represents a dimensional reduction that cannot be fully tested from within. The appropriate response is not to abandon empirical inquiry but to hold frameworks as tools rather than truths, to expect scope limitations, and to remain alert for situations where framework choice dominates over empirical content.

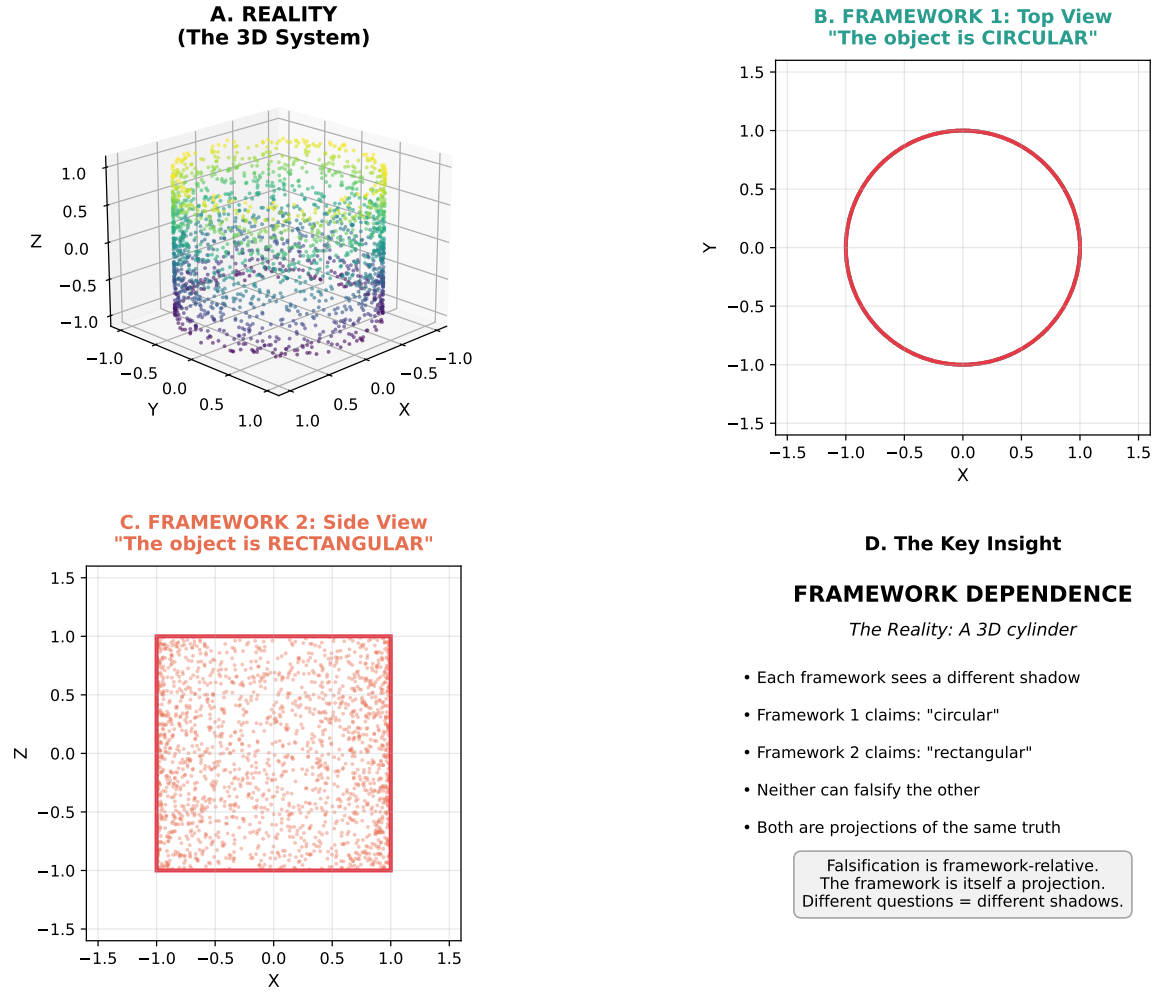


Figure 2: **Framework as Projection.** A cylinder (A) casts different shadows depending on projection angle. Framework 1 (B) observes only the XY projection and concludes the object is circular. Framework 2 (C) observes only the XZ projection and concludes it is rectangular. Both projections are “correct” given their assumptions, but they generate incompatible hypotheses. Neither can falsify the other *without adopting a shared higher-dimensional measurement* because they are not making claims in the same dimensional subspace.

3 The Binary Projection Problem in Biology

3.1 Implicit Assumptions in Biological Falsifiability

Scientific demarcation practice typically induces a decision boundary, even when measurement and inference are continuous. While Popper concerned himself with *in principle* falsifiability, we show that physical laws impose *in principle* limits on what biological phenomena can be falsified.

Clarification on Popper and binary tests. A skeptical reader may object that Popper’s falsifiability is a *logical relation* between hypothesis and observation—observations can be continuous, noisy, statistical—and that modern “Popperian” practice often uses statistical falsification (likelihood ratios, posterior predictive checks) rather than literal single-bit measurements. We accept this clarification but note it does not dissolve our argument. The epistemic bottleneck is not in the measurement device but in the *demarcation logic*: even if raw measurements are rich, the scientific process ultimately forces a decision boundary (accept/reject, significant/non-significant, model A/model B). Even when conclusions are graded (effect sizes, posterior odds), *action*—accept a model, fund a project, approve a drug, revise a theory—still induces a boundary. It is this forced projection from continuous evidence to discrete conclusion that creates information loss, regardless of how sophisticated the intermediate analysis.

Bayesianism does not escape. One might hope that Bayesian inference—which maintains full posterior distributions rather than forcing binary decisions—circumvents this problem. In the regimes we consider (high dimensionality, weak or uninformative priors, limited computational budget), it does not. Bayesian methods face their own high-dimensional crises: the curse of dimensionality often makes posterior computation intractable, priors become effectively uninformative or encode unjustified assumptions, MCMC samplers frequently fail to mix, and variational approximations introduce uncontrolled bias. More fundamentally: if the posterior cannot be computed or even represented, Bayesian analysis cannot begin. The

promise of “maintaining full distributions” becomes vacuous when the distribution lives in a space too large to explore. In high-dimensional biological systems with non-identifiability and limited data, Bayesian inference does not preserve information—it projects onto a tractable subspace, just as frequentist methods do. Moreover, even with infinite computational resources, Bayesian inference is limited by the *support of the prior*: if the hypothesis space itself is a low-dimensional projection of the system’s true degrees of freedom, the posterior will converge on the “best fit” within that projection, not the truth. The framework dependence runs deeper than the choice of statistical paradigm.

Consider a hypothesis about protein folding: “Protein X folds via pathway Y.” Testing this proposition requires bringing the protein to a state where folding intermediates are discretely resolvable, measuring without disrupting the folding process, and projecting the multi-dimensional folding landscape onto a binary decision axis. Yet protein folding occurs on a rugged energy landscape with astronomical numbers of conformational states [7]. The act of measurement necessarily perturbs this landscape, potentially switching the protein to alternative folding pathways. Moreover, the binary projection destroys information about parallel pathways, transient intermediates, and the inherently statistical nature of the folding process.

3.2 Information Loss Under Projection

For a biological system with n degrees of freedom, each with k distinguishable states, the *state specification information* (microstate description length) is:

$$I_{\text{spec}} = \log_2(k^n) = n \log_2 k \quad \text{bits} \tag{1}$$

A binary test extracts exactly 1 bit—a yes/no partition of the state space. The fraction

of information preserved is therefore:

$$\frac{I_{\text{preserved}}}{I_{\text{spec}}} = \frac{1}{n \log_2 k} \quad (2)$$

In a modest neural circuit with $n = 100$ neurons, each with $k = 10$ distinguishable states (an illustrative value; actual neural state spaces may be larger), this ratio becomes $1/(100 \times 3.32) \approx 0.003$ —less than 1% of the state specification information preserved. A single binary test discards over 99% of the system’s microstate description length.

Optimized projections fare little better. One might object that scientific questions are not random partitions but *directed* projections—like principal component analysis (PCA)—designed to capture maximum variance, and that the “relevant” information may live on a low-dimensional manifold. This is true, but it does not rescue the situation. Even optimized projections must discard the “long tail” of high-dimensional correlations, and in biological systems this tail often contains the critical causal structure: rare conformational states, weak but decisive couplings, transient coherences that gate function. The principal components capture what varies most, not what matters most. Optimizing the projection does not eliminate projection loss; it merely concentrates the preserved information in directions chosen by the optimizer’s assumptions—another layer of framework dependence.

Lemma 1 (Identification is framework-dependent). *Even if the relevant structure is low-dimensional, identifying that structure is itself a high-dimensional inference problem. There is no framework-free way to determine which low-dimensional projection is “correct.”*

Proof. See Appendix A. □

This is not a practical limitation awaiting better technology. It is an information-theoretic fact about projection. Moreover, the situation compounds: to fully specify the system state would require $n \log_2 k \approx 332$ independent binary tests, each of which may perturb the system. The measurement problem is not just that individual tests are lossy, but that the number of tests required scales with system complexity while biological systems cannot tolerate

arbitrary repeated measurement.

4 Three Axes of Measurement Limitation

Beyond framework dependence, three orthogonal physical constraints limit falsifiability:

4.1 High Dimensionality

Biological systems are fundamentally high-dimensional. A single cell’s state requires thousands of variables to specify—gene expression levels, metabolite concentrations, protein conformations, membrane potentials. Any finite-dimensional description is a projection.

Any concrete mathematical model has finite parameterization: one can only write finitely many symbols, equations, and variables. Every model is therefore a dimensional reduction. The question is not whether projection occurs but whether projection loss is small enough to preserve relevant structure. For biology, it often is not.

4.2 Quantum Timing and Measurement Disturbance

Even for low-dimensional systems, quantum measurement backaction imposes limits. When biological processes depend on coherent superposition or entanglement (as in photosynthetic energy transfer [8]), measurement to resolve the system state necessarily collapses the superposition, destroying the very coherence that enables function.

Unlike position or momentum, time in standard quantum mechanics is a parameter rather than an observable; time-of-arrival is accessible only as a distribution, and precise timing readouts introduce back-action via non-commuting constraints. When such micro-timings are chaos-amplified, complete specification is limited in principle.

4.3 Thermodynamic Erasure and the Landauer Bound

At physiological temperature ($T \approx 310$ K), the Landauer limit for irreversible bit erasure is:

$$E_{\text{Landauer}} = k_B T \ln 2 \approx 3.0 \times 10^{-21} \text{ J} \quad (3)$$

Strictly, Landauer’s principle concerns the thermodynamic cost of *erasing* information, not detecting it. However, any measurement that produces a reusable, communicable result must eventually involve erasure—resetting the measurement apparatus for subsequent use. The deeper issue is *timing inaccessibility*: even if detection were thermodynamically free, specifying *when* to measure in a chaotic system requires information that cannot be obtained without perturbing the dynamics [25].

ATP hydrolysis releases approximately 5×10^{-20} J, only 17 times the Landauer limit. Many biological signals operate near or below this fundamental limit. Sub-threshold membrane fluctuations, protein conformational vibrations, and weak molecular interactions fall in this regime, where reliable binary readout becomes physically constrained.

These three axes—dimensionality, quantum disturbance, and energetic erasure—often compound: high-dimensional biological systems operating near thermodynamic limits with quantum-like coherence represent the regime where falsifiability breaks down most completely.

5 The Sub-Landauer Domain

5.1 Definition

We define the sub-Landauer domain as the space of biological patterns whose energy content falls below reliable single-event readout energy:

Definition. A biological structure $\mathcal{P}$ is *sub-Landauer* if:

1. Its energy scale $E_{\mathcal{P}} \lesssim k_B T \ln 2$ (order-of-magnitude condition)

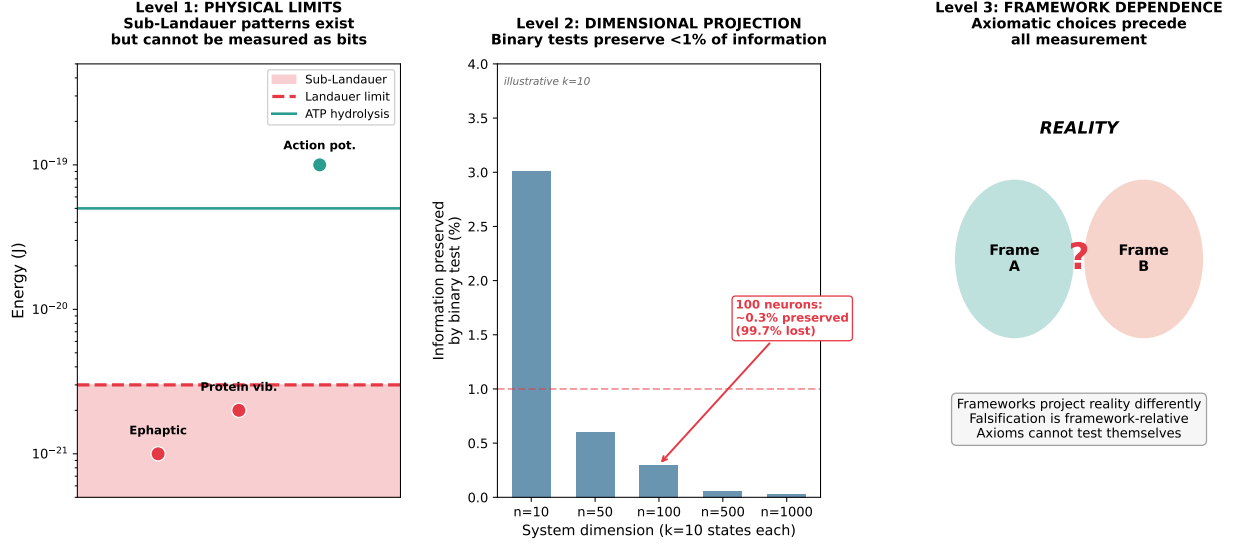


Figure 3: **Three Levels of Limitation.** Left: Physical measurement limits—many biological signals exist below reliable single-event discrimination energy. Center: Dimensional projection loss—a binary test on a 100-neuron circuit preserves less than 1% of the information content ($k = 10$ states per neuron is illustrative; actual neural state spaces may be larger). Right: Framework dependence—axiomatic choices about what counts as evidence precede all measurement and cannot be tested from within the framework.

2. It exhibits temporal coherence beyond thermal relaxation
3. It causally influences observable biological functions
4. Measurement sufficient to resolve it as a bit destroys the role *by perturbing the dynamics beyond the relevant tolerance*—e.g., disrupting coherence, altering pathway selection, or shifting phase relations

Note on terminology. The Landauer limit ($k_B T \ln 2 \approx 3 \times 10^{-21}$ J at physiological temperature) is a convenient reference scale; the substantive constraint is reliability against thermal fluctuations for single-event discrimination. Patterns at or below this scale cannot be reliably read as individual bits, though they remain causally potent and collectively detectable.

Operationalizing E_P . For patterns involving fields, correlations, or distributed codes, the “energy scale” requires operationalization. We define E_P as the *energy difference per*

effective degree of freedom per correlation time between the pattern-present and pattern-absent states. For an electric field perturbation, this is $\frac{1}{2}\epsilon E^2 \cdot V/D_{\text{eff}}$ where V is the affected volume and D_{eff} the number of coherently coupled modes. For a population correlation pattern, it is the change in free energy when the correlation is imposed versus destroyed. In each biological example below, we indicate how this quantity is estimated.

5.2 Biological Examples

Photosynthetic energy transfer. Quantum coherence in photosynthetic complexes exists at energies near 10^{-21} J—near the Landauer limit. The coherence enables near-perfect energy transfer efficiency by allowing excitons to simultaneously sample multiple pathways. Measurement to determine the specific path destroys the coherence and reduces transfer efficiency [8, 10].

Ephaptic coupling. Endogenous electric fields as weak as 1 mV/mm—well below single neuron detection thresholds—can shift spike timing by several milliseconds across neural populations [9, 11]. These fields represent information-carrying patterns that operate below the Landauer limit yet causally influence network dynamics.

Quantitative estimate. Consider a voltage-gated ion channel—the fundamental “decision unit” of neuronal excitability—responding to an ephaptic field. At body temperature ($T \approx 310$ K), the Landauer limit is $E_{\text{Landauer}} = k_B T \ln 2 \approx 3.0 \times 10^{-21}$ J. An extracellular field of 1 mV/mm induces transmembrane polarization of approximately $\Delta V \approx 0.1$ mV. The voltage sensor of a typical Na^+ or K^+ channel has effective gating charge $q \approx 4e$ (where $e = 1.6 \times 10^{-19}$ C). The work done by the ephaptic perturbation on this gating charge is:

$$W = q\Delta V \approx (4 \times 1.6 \times 10^{-19} \text{ C})(10^{-4} \text{ V}) \approx 6.4 \times 10^{-23} \text{ J} \quad (4)$$

This is approximately **2% of the Landauer limit**. (This is a *work-scale proxy*, not the full energetic budget of sensing; it bounds how much the field can bias gating per channel

per event.) A binary detector at the single-channel level would read only thermal noise. Yet these fields successfully entrain network activity because collective integration across N channels lowers the effective threshold by $\sqrt{N}$ (assuming approximately independent noise across units over the correlation time)—the signature of stochastic resonance operating in the sub-Landauer regime.

Stochastic resonance. Weak periodic signals below individual neuron thresholds can entrain population activity through noise-mediated synchronization [6, 13]. The effective detection threshold drops as $E_{\text{threshold}}/\sqrt{N}$ for N coupled units, rendering sub-threshold drives decisive at the population level despite being individually unmeasurable.

5.3 Collective Computation Through Stochastic Resonance

Biological systems exploit stochastic resonance to amplify sub-Landauer signals. In neural populations, weak periodic signals below individual neuron thresholds can entrain population activity through noise-mediated synchronization [14]. The mechanism requires:

$$E_{\text{signal}} + E_{\text{noise}} > E_{\text{threshold}} \quad (5)$$

where individually $E_{\text{signal}} < E_{\text{Landauer}}$ but collectively the signal emerges through correlation across many units.

Let $s(t)$ be a weak periodic drive with per-unit energy $E_{\text{signal}} < E_{\text{Landauer}}$ and independent noise $\eta_i(t)$ at units $i \in \{1, \dots, N\}$. For $Y(t) = \frac{1}{N} \sum_i y_i(t)$ with y_i a thresholded response,

$$\text{SNR}(Y) \approx \frac{N \text{cov}(y_i, s)}{\sqrt{N \text{var}(y_i)}} \propto \sqrt{N} \quad (6)$$

so the effective detection threshold scales as $E_{\text{threshold}}/\sqrt{N}$ (under the same independence/correlation-time assumptions as in §5.2). This is suprathreshold stochastic resonance [13], observed in sensory systems and neural arrays [15, 16].

The key insight is that sub-Landauer patterns are *decisive* for biological function precisely because they operate at criticality—the boundary between noise and signal.

Clarification on the “bit.” The Landauer limit applies to recording a single bit irreversibly. When we say sub-Landauer signals are “unmeasurable,” we mean that no *single* event carries enough energy to flip a reliable detector. But ensembles of such events can be detected: the “bit” extracted is a statistical property of the population (e.g., the mean phase), not any individual sub-threshold event. This is precisely how stochastic resonance works—the signal emerges from correlation across many units, none of which individually crosses threshold. The ensemble can be measured; the individual events constituting it cannot.

6 Analytical Bounds on Predictability

6.1 The Specification Horizon

To formalize the limits on complete system specification, consider a biological dynamical system with n coupled degrees of freedom, maximum Lyapunov exponent λ , and quantum uncertainty $\Delta x_i \geq \hbar/(2\Delta p_i)$ in each coordinate. The predictability horizon T_{pred} for deterministic specification is bounded by:

$$T_{\text{pred}} \lesssim \frac{1}{\lambda} \ln \left(\frac{L}{\Delta x} \right) \quad (7)$$

where L is the characteristic dynamic range of the relevant coordinate(s) and Δx is the *effective* state-estimation precision under the chosen measurement map M , including measurement noise and any backaction induced by acquiring that precision (cf. §2.2). This ties the predictability bound back to framework dependence: different choices of M yield different Δx and hence different horizons. This is an *in-principle* bound: specification cannot be extended beyond this horizon without changing the system dynamics or the functional

regime being specified.

6.2 Information-Theoretic Requirement

For n effective degrees of freedom resolved to precision Δx over a domain of size L , the binary information required for full specification scales as:

$$I_{\text{required}} \sim \frac{n}{\ln 2} \ln \left(\frac{L}{\Delta x} \right) \quad \text{bits} \quad (8)$$

When I_{required} exceeds the maximum extractable information without destroying function, single-trial binary falsification becomes incoherent; only ensemble fingerprints remain accessible. This formalism shows that computational irreducibility [17] is not merely a practical limitation but follows from fundamental physical constraints.

7 Emergent Properties and Collective Phenomena

Biological systems exhibit emergent properties that exist only at the collective level. Swarming behavior in bacteria emerges from quorum sensing—a phenomenon where individual cells respond to population density through diffusible signals [18]. The swarm’s properties cannot be reduced to binary statements about individual cells.

Similarly, consciousness appears to emerge from integrated information across brain regions [12]. Attempts to falsify theories of consciousness through binary tests necessarily destroy the integration that defines the phenomenon.

Evolution operates in fitness landscapes of enormous dimensionality. A single protein’s fitness depends on its interactions with thousands of other molecules, environmental conditions, and evolutionary history. Wright’s adaptive landscape metaphor [19], while useful, dramatically understates the true dimensionality—real fitness landscapes may have millions of dimensions.

8 Epistemological Implications

8.1 The Conjunction of Limits

The three levels of limitation—framework dependence, dimensional projection, and physical measurement bounds—interact synergistically. Even if measurement were perfect, dimensional projection would destroy information. Even if dimensionality were low, framework dependence would make falsification relative to unstated assumptions. Together, they circumscribe a domain where classical falsifiability becomes incoherent.

This domain is not marginal. It includes:

- Consciousness and neural integration
- Protein folding and molecular recognition
- Evolutionary dynamics on fitness landscapes
- Ecological network stability
- Developmental morphogenesis

These are not failures of current methodology awaiting future resolution. They represent fundamental limits on what binary epistemology can access.

8.2 Why Disagreement Persists

The framework-dependence of falsifiability explains why disagreement persists in biology despite good-faith empirical investigation. Researchers in different paradigms are not making claims in the same framework. Their hypotheses project differently. What counts as evidence differs. The “same” experiment can support different conclusions depending on auxiliary assumptions.

This is not irrationality. It is the geometric consequence of framework-relative inquiry. Resolution requires not more data but framework negotiation—explicit discussion of what assumptions are being made and whether they are shared.

8.3 Implications for Consciousness (Speculative)

Note: This subsection is more speculative than the preceding analysis. The arguments about measurement limits and framework dependence are established; their application to consciousness is a *hypothesis* rather than a derived conclusion. Alternative explanations for the “hard problem” exist and are not precluded by our framework.

The hard problem of consciousness may be hard precisely because consciousness emerges from high-dimensional, sub-Landauer coherence patterns. If conscious experience depends on patterns that:

1. Exist below measurement thresholds
2. Require integration across distributed neural populations
3. Are destroyed by the dimensional reduction of measurement

then consciousness is in principle inaccessible to binary falsification. The phenomenon would be real and causally efficacious yet invisible to third-person methodology.

This does not make consciousness mystical. It makes it a different kind of thing than what physics typically describes—not because it violates physical laws, but because it is constituted by dynamics that physical measurement cannot fully access.

9 Toward Multi-Scale Epistemology

9.1 Scale-Dependent Falsifiability

We propose that falsifiability in biology is scale-dependent:

Principle (Scale-Dependent Falsifiability). A biological hypothesis H about system S is falsifiable at energy scale E and framework $\mathcal{F}$ if:

$$E \gtrsim \max(E_{\text{Landauer}}, E_{\text{coherence}}(S), E_{\text{coupling}}(S))$$

and H is expressible within the shared axiomatic structure of $\mathcal{F}$.

Here $E_{\text{coherence}}(S)$ is the order-of-magnitude energy required to resolve coherent patterns in S without destroying them (e.g., the energy scale of phase-locking or quantum superposition), and $E_{\text{coupling}}(S)$ is the energy required to perturb the couplings that constitute S 's functional organization. The criterion uses $\gtrsim$ (order-of-magnitude inequality) rather than strict $>$, consistent with the sub-Landauer definition in §5.1.

This acknowledges that some biological questions admit falsification (enzyme kinetics, action potential propagation) while others do not (consciousness, ecosystem stability), and that even falsifiable hypotheses are only falsifiable relative to a shared framework.

9.2 Alternative Validation Approaches

Given these limitations, biological sciences require validation methods beyond falsification:

- **Pattern consistency:** Test whether models reproduce statistical patterns across scales
- **Predictive power:** Evaluate probabilistic predictions over ensembles
- **Mechanistic coherence:** Assess consistency with physical and chemical constraints
- **Convergent evidence:** Integrate multiple indirect lines of support
- **Framework transparency:** Explicitly state axiomatic assumptions

No single approach suffices. Biological validation requires a portfolio of complementary methods that acknowledge framework dependence and scale limitations.

Example: Ensemble-based inference in neural coding. Consider testing whether a neural population encodes a particular stimulus feature. A single-trial binary test (“did neuron i fire?”) preserves almost no information about the population state. But measuring firing rates across many trials, computing cross-correlations, or extracting low-dimensional manifold structure from high-dimensional spike trains—these ensemble methods can reveal coding principles that single-trial falsification cannot access. The “hypothesis” is not “neuron i fires when stimulus X is present” but rather “the population trajectory occupies region R of state space with probability $p > p_0$ when X is present.” This is falsifiable, but only statistically, and only with respect to a framework that defines the relevant state space, the appropriate dimensionality reduction, and the threshold p_0 . The framework dependence is explicit, not hidden.

10 Conclusion

Karl Popper’s falsifiability criterion has been invaluable for biology, helping distinguish science from pseudoscience. However, our analysis reveals three levels of limitation:

1. Physical measurement constraints create a sub-Landauer domain of causally potent but unmeasurable patterns
2. Dimensional projection destroys almost all information when high-dimensional systems are reduced to binary tests
3. Framework dependence makes all falsification relative to unstated axiomatic assumptions

The “unreasonable effectiveness of mathematics” in physics reflects selection bias: physics has focused on domains where projection loss is small and frameworks are shared. Biology is where this selection breaks down.

This is not a failure of scientific methodology but a recognition of fundamental limits on biological knowledge. The future of biological science lies not in universal falsifiability but in a scale-aware, dimension-sensitive, framework-transparent epistemology that respects the limits physics places on what can be known about living systems.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

This is Version 2.0 of a paper originally published in BioSystems (2025). The original version was developed with Claude 4 (Anthropic). This upgraded version was developed with Claude 4.5 Opus (Anthropic), incorporating new arguments about framework dependence and mathematics as projection. The author reviewed and edited all content and takes full responsibility for the content.

Version History

- **v1.0** (October 2025): Published in BioSystems. DOI: 10.1016/j.biosystems.2025.105608

- **v2.0** (December 2025): Expanded with framework-dependence argument, Wigner/selection-bias analysis, and mathematics-as-projection thesis. Available at: <https://coherencedynamics.com/papers/falsifiability>

A Proof of Lemma 1

Lemma 1 (Identification is framework-dependent). *Even if the relevant structure is low-dimensional, identifying that structure is itself a high-dimensional inference problem. There is no framework-free way to determine which low-dimensional projection is “correct.”*

We proceed by showing that any identification procedure necessarily constitutes a framework in the sense of Definition 1 (§2.2), and that the choice of framework cannot be justified without appeal to a further framework.

Step 1: Identification requires algorithmic choices. Let X be a high-dimensional state space and suppose the “relevant” structure lies on a low-dimensional submanifold $\mathcal{M} \subset X$. Identifying $\mathcal{M}$ from observations requires an *algorithm*—a finite sequence of operations mapping data to a specification of $\mathcal{M}$.

Any such algorithm must make the following choices:

1. **Feature selection:** Which observables $Y = M(X)$ to measure? This choice determines what aspects of X are visible.
2. **Representation:** How to encode observations into a tractable form $Z = R(Y)$? This includes choices of basis, normalization, and preprocessing.
3. **Criterion:** What constitutes a “good” identification? This requires a loss function, likelihood, or selection criterion.
4. **Regularization/prior:** How to handle underdetermination when data does not uniquely specify $\mathcal{M}$?

Step 2: These choices constitute a framework. The triple (M, R, D) —where D is the decision rule induced by the criterion—defines a framework $\mathcal{F}$ in the sense of §2.2. The framework maps the state space X through measurement M , representation R , and decision D to produce an identification output.

Each framework $\mathcal{F}$ induces a partition $\{P_i\}$ on X : states in the same partition element are indistinguishable under $\mathcal{F}$. Different frameworks induce different partitions.

Step 3: Framework choice cannot be justified framework-free. Suppose we attempt to adjudicate between frameworks $\mathcal{F}_1$ and $\mathcal{F}_2$ to determine which “correctly” identifies the relevant structure. This adjudication requires:

- A criterion for what counts as “correct” identification
- A method for comparing framework outputs
- Data to inform the comparison

But these requirements themselves constitute a meta-framework $\mathcal{F}^*$. The choice of $\mathcal{F}^*$ faces the same problem: it cannot be justified without appeal to some further framework $\mathcal{F}^{**}$. The regress does not terminate.

Step 4: Conclusion. Since (i) every identification procedure constitutes a framework, (ii) different frameworks identify different structures (induce different partitions), and (iii) framework choice cannot be justified without infinite regress, identification is framework-dependent.

This holds even when the “true” low-dimensional structure exists objectively. The structure’s existence does not guarantee framework-free access to it. The map from high-dimensional observations to low-dimensional identification is necessarily mediated by choices that cannot be eliminated or uniquely determined by the data.

B Information-Theoretic Formalization

The framework dependence of identification can be stated information-theoretically using the Data Processing Inequality (DPI).

Setup. Let X be the true system state (high-dimensional), $Y = M(X)$ the measurement, $Z = R(Y)$ the representation, and $\hat{\mathcal{M}} = D(Z)$ the identified structure.

Proposition (Projection loss is unavoidable). For any framework $\mathcal{F} = (M, R, D)$:

$$I(X; \hat{\mathcal{M}}) \leq I(X; Z) \leq I(X; Y) \leq H(X) \quad (9)$$

where $I(\cdot; \cdot)$ denotes mutual information and $H(\cdot)$ denotes entropy. The inequalities are strict whenever the maps are non-invertible.

Proof. By DPI, post-processing cannot increase information: $X \rightarrow Y \rightarrow Z \rightarrow \hat{\mathcal{M}}$ forms a Markov chain, so $I(X; \hat{\mathcal{M}}) \leq I(X; Z) \leq I(X; Y)$. The bound $I(X; Y) \leq H(X)$ follows from the definition of mutual information. $\square$

Corollary. Different frameworks $\mathcal{F}_1, \mathcal{F}_2$ preserve different information about X . Specifically, $I(X; \hat{\mathcal{M}}_1)$ and $I(X; \hat{\mathcal{M}}_2)$ depend on the framework-specific maps, and in general neither dominates the other. The “correct” framework cannot be determined by maximizing preserved information without first specifying *which* information matters—another framework choice.

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